

The Depreciation of Understanding: Continuity as the Missing Variable in Endogenous Growth Theory

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“In this inquiry, we argue that the economic value of an idea is jointly determined by its existence and by the continuity of coherent understanding required to access it — and that these two variables, contrary to the implicit assumptions of endogenous growth theory, can move independently of each other.” — Arne Mayoh

Abstract

Endogenous growth theory identifies ideas as the central engine of long-run economic growth, distinguished from physical capital by non-rivalry: one person’s use of an idea does not diminish another’s, and — critically — ideas, once discovered, do not depreciate. This paper argues that non-rivalry of existence is not the same as non-depreciation of value. Drawing on this series’ account of measurement conventions as prior theoretical commitments (Paper 1), navigational validity for open systems (Paper 2), Interpretive Entropy as the mechanism by which reasoning coherence degrades in fragmented systems (Paper 3), the non-linear threshold dynamics of that degradation (Paper 4), the resulting unmeasured economic cost (Paper 5), and understanding as a sustained field-relationship rather than a possessed object (Paper 6), this paper proposes that the *coherent relationship* to an idea — not the idea’s mere persistence in some record — is the actual unit of durable economic value, and that this relationship depreciates independently of the idea itself. What follows names this relationship, in economic terms, continuity capital — the same underlying phenomenon this series’ companion volume calls continuity of understanding, addressed here in the register its measurement requires. The implication is a previously unnamed asset class.

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Companion publication: *The Wealth of Understanding* (Arne Mayoh, forthcoming) — this paper formalizes the claim the book states only as hook and conclusion (Ch.8’s “coherence compounds” passage)

I. A Recipe That Was Never Lost

In roughly 25 BCE, the Roman architect Vitruvius wrote down, in *De Architectura*, a description of how to make concrete using lime, volcanic ash, and seawater. The text survived. It was copied, translated, printed, and eventually digitized. For two thousand years, anyone literate in Latin — and for the last several centuries, anyone with a library card — could read exactly what the Romans claimed to have done.

The idea, in Paul Romer’s precise sense, never disappeared. It was never rivalrous, never destroyed, never even damaged. One reader’s access to Vitruvius’s description did nothing to diminish another’s. By the standard account of what makes an idea economically valuable — that it exists, and that its existence is non-rival — the recipe for Roman concrete has been sitting in plain view, freely available to the entire literate world, for the whole of recorded modern history.

And yet, for essentially all of that history, the recipe was economically worthless. Modern engineers who followed Vitruvius’s instructions did not produce Roman concrete. They produced concrete that eroded, cracked, and failed in decades where the originals had endured for two millennia — the sea walls and harbor structures Roman builders submerged directly in the Mediterranean, still standing today. The gap between having the idea and having the *use* of the idea was not closed by the idea’s continued existence. It was closed only by three separate teams of scientists, working across more than a decade and two continents, who reconstructed — not read, reconstructed — the actual mechanism the Romans had relied on: Jackson and colleagues’ identification of aluminous tobermorite crystallizing from seawater interaction (2013), the same team’s account of the mortar’s mechanical resilience (2014), and, a decade later, Seymour, Masic, and colleagues’ discovery that the Romans were using a “hot-mixing” process with quicklime that produced reactive lime clasts capable of self-healing cracks as they formed (2023). Each finding corrected or extended the last. None of them was in Vitruvius. All of them were true of what Roman builders actually did.

The recipe was never lost. The understanding was.

This paper is about the space between those two sentences.

Endogenous growth theory, in its most influential form, treats an idea’s economic contribution as a function of its existence: once discovered, non-rival, permanently available, doing its work indefinitely (Romer, 1990). This is one of the genuinely load-bearing results of modern economics, and nothing that follows disputes it as an account of what makes ideas different from physical capital. But the Roman concrete case suggests the model is silent about a variable that turns out to matter enormously: whether the *coherent relationship* to an idea — the operational, reconstructable understanding of why it works and how to reproduce it — persists alongside the idea’s bare existence, or depreciates independently of it. Vitruvius’s text was non-rival the entire time. Its economic value was, for most of two thousand years, zero. Those two facts are not in tension under Romer’s model. They are not supposed to happen at the same time.

The remainder of this paper argues that they happen together far more often than growth theory currently accounts for; that this series’ prior work on measurement conventions, navi-

gational validity, interpretive entropy, and continuity thresholds already supplies the mechanism by which it happens, and that the resulting asset — the coherent, maintained relationship to an idea, as distinct from the idea itself — deserves to be named and measured on its own terms.

A closer relative still is Paul Romer’s account of endogenous growth. Romer’s insight — that ideas, unlike rival physical goods, can be used without being consumed, and therefore permit growth unbounded by the scarcity that constrains capital and labor — is one of the genuine load-bearing results of modern growth theory, and nothing in this paper disputes it. It is worth being precise about where this argument agrees with Romer and where it departs. Romer’s model treats an idea as a binary state: discovered or not. Once discovered, it is simply *there*, permanently available, doing its non-rival work indefinitely. That presupposition holds exactly to the extent that an idea’s usefulness depends only on its existence and not on anyone’s continued, coherent relationship to *why* it works, *what* it assumes, and *how* it was arrived at. It does not hold for the specific class of cases this series has documented: where the reasoning behind an idea degrades, fragments, or must be expensively reconstructed even while the idea’s symbolic trace persists untouched in a record. This paper does not compete with endogenous growth theory on the ground it already occupies. It proposes that the ground it does not occupy — the maintenance cost of coherent access to an idea already discovered — is where a comparable share of long-run economic value is actually being won or lost.

II. The Claim, Stated Precisely

This paper advances a single claim, stated as narrowly as it can be while still being worth stating:

The economic value of an idea is jointly determined by its existence and by the continuity of coherent understanding required to access it — and these two variables can move independently of each other.

Three implications follow directly.

First: non-rivalry of existence, Romer’s contribution, remains true and remains important. This paper does not propose that ideas depreciate in the sense that using them wears them out. Vitruvius’s text is exactly as intact today as it was in 1500.

Second: a second variable, absent from the growth-theory production function, determines whether that non-rival existence converts into usable value. Call it continuity — following this series’ usage — meaning the maintained, reconstructable coherence between an idea and the reasoning that generated it. Continuity is not a property of the idea. It is a property of the *relationship* a community of practice maintains to the idea, and it can degrade while the idea itself is preserved perfectly.

Third: because continuity is a relationship rather than an object, it obeys different dynamics than the non-depreciating ideas in Romer’s model. It can be lost gradually, lost suddenly past a threshold, or in principle rebuilt at a reconstruction cost that may exceed the value of rebuilding it at all. Each of these dynamics has already been established, independently, in this series, for reasoning generally. This paper’s contribution is to argue that they apply, specifically and

consequentially, to the economics of ideas — the exact object Romer’s model treats as immune to them.

A note on terminology, stated explicitly rather than left implicit. What this paper calls continuity capital is the economic name for what this series’ companion volume, and Paper 6, call continuity of understanding. The substitution is deliberate and is disclosed here rather than allowed to happen silently across the paper. Understanding, as a term, carries a philosophical weight and a contested definitional history — the object of long-running disagreement among philosophers of understanding — that this paper does not need to resolve and is not equipped to adjudicate. Paper 6 already made the relevant move for this series: understanding is not a possessed object but a maintained field-relationship, sustained in tension rather than resolved into a fixed state. This paper takes that result as given and asks a narrower, economic question about it: if understanding is a maintained relationship rather than a possessed object, does that relationship have economic value distinct from the ideas it attaches to, and does that value depreciate on its own terms? Continuity capital is the name for that value, chosen specifically because it names the relationship — the thing that can depreciate — rather than the experience or philosophical category of understanding itself, which cannot be an object of economic measurement in the same way. The two vocabularies are not competing claims. They are the same underlying claim, in Paper 6’s philosophical register and this paper’s economic one, and the choice to use the more technical term here should be read as a decision about audience and unit of analysis, not as a retreat from the claim that understanding — under its more precise name — is what is actually at stake. Naming this substitution explicitly is intended to invite continued dialogue across registers — economic, governance, and philosophical — as a single systemic conversation about the same phenomenon, rather than allowing the vocabulary shift to quietly wall the economic argument off from the philosophical one it depends on.

III. What This Series Has Already Established

The argument does not require new mechanisms. It requires recognizing that mechanisms already established for reasoning in general apply, as a special case, to the ideas that endogenous growth theory treats as the engine of economic growth.

Measurement conventions are prior theoretical commitments, not neutral observations (Paper 1). GDP does not fail to capture the depreciation of continuity by oversight. It was built to measure market production in a physical-capital economy, and a convention built to see one thing is not thereby equipped to see another. The invisibility of continuity depreciation in growth accounting is not a data problem. It is the design condition Paper 1 describes: the phenomenon was never a candidate for measurement because the convention was never built with it in view.

Navigational validity is the appropriate standard for claims about open, adaptive systems (Paper 2). The claim advanced here cannot be tested by isolating an idea and observing its depreciation under laboratory control, for the same reason Paper 2 identifies for complex systems generally: isolating the idea from the community of practice that maintains coherent access to it changes the very thing being measured. What follows is offered as a navigational

hypothesis, in Paper 2's sense — specifying an expected pattern in advance, improving orientation among competing accounts of the productivity paradox, and specifying its own conditions for revision — not as a claim awaiting a controlled experiment that its own subject matter forecloses.

Interpretive Entropy names the mechanism of degradation (Paper 3). This is the paper's central borrowed engine. Interpretive Entropy describes reasoning coherence degrading as it is reconstructed, from fragments, by parties who lack access to the original interpretive context — a process distinct from information loss, since the symbols can survive perfectly while the meaning does not. Applied to Vitruvius: the words survived. The interpretive context — why quicklime, why this ratio, why seawater curing produced strength rather than erosion — did not travel with them, because it was never made explicit in the first place. It lived in practice, in the bodies and workshops of Roman builders, and when that practice ended, the coherent relationship to the written idea ended with it. The idea did not depreciate. The community's interpretive access to it did.

Continuity Threshold dynamics make this a non-linear, not gradual, process (Paper 4). Growth theory, to the extent it considers depreciation at all, models it as smooth decay. Paper 4's account of hysteresis — that recovery past a threshold requires more favorable conditions than preceded the crossing — offers a sharper and more falsifiable prediction: continuity depreciation should show threshold behavior, not smooth decay, and recovery cost should be asymmetric to loss cost. The multi-decade, multi-institution effort required to partially reconstruct Roman concrete's mechanism, against the near-zero cost of simply reading Vitruvius, is consistent with exactly this asymmetry, though a single historical case is illustrative rather than confirmatory.

The resulting cost is real, economically significant, and currently unmeasured (Paper 5). This paper is the direct ancestor of the present claim; what follows generalizes Paper 5's specific accounting category into a claim about the underlying asset whose loss that category was measuring the cost of.

Understanding is a sustained field-relationship, not a possessed object (Paper 6). This is the paper's deepest dependency, and the one Romer's model implicitly forecloses by treating an idea as a discrete, binary state — discovered or not. Paper 6's argument that meaning exists in the maintained tension between poles, not in either pole alone, is the philosophical warrant for saying that an idea's *usefulness* is not a property of the idea in isolation at all. It is a property of the field — the sustained, coherent relationship between the idea and the community that can still activate it. Vitruvius's text, absent that field, is not a diminished idea. It is, functionally, closer to what Paper 6 calls incoherence: information present, relationship absent.

Taken together, these six papers do not need to be extended to reach the claim in Section II. They need only to be pointed, jointly, at a target — the economics of ideas — that none of them individually addressed but all of them jointly equip this paper to address.

IV. Continuity Capital: A Proposed Asset Class

If continuity is a variable independent of an idea's existence, and if its depreciation follows the mechanisms established in Papers 3 and 4, then it is reasonable to treat continuity itself

as a form of capital — an asset that can be invested in, allowed to depreciate, or lost entirely, distinct from the idea it attaches to. This paper proposes the term **continuity capital**: the accumulated, maintained coherence between an idea and the community of practice capable of reconstructing why it works, sufficient to make the idea’s existence economically activable rather than merely available.

Continuity capital is not a restatement of human capital or intellectual property. Human capital resides in individuals; continuity capital resides in the relationship between an idea and a community, and can outlive any individual member of that community or collapse despite every individual remaining alive and informed, if the relationship among them fragments. Intellectual property protects the right to exploit an idea; it says nothing about whether anyone remaining can still exploit it competently. Vitruvius’s recipe was never encumbered by IP in any sense that would explain its two-thousand-year economic dormancy. What it lacked was continuity capital: a maintained community whose coherent, reconstructable understanding of the underlying mechanism made the recipe usable rather than merely legible.

This is a substantial claim, and Paper 2 established the standard this series holds itself to before making claims of this kind. The claim is tested against that standard directly, condition by condition, rather than asserted to deserve it.

Condition 1 — the phenomenon cannot be isolated without changing it. Continuity capital cannot be studied by removing an idea from its community of practice and measuring what remains, because the community’s coherent relationship to the idea *is* the phenomenon; isolating the idea for controlled study removes the object of inquiry along with its context, exactly as Paper 2 specifies for genuinely non-isolable phenomena. This is not a limitation to be engineered around. It is why the Roman concrete case is illustrative rather than experimental: the depreciation had already occurred, over centuries, before anyone thought to measure it, and no controlled trial could have run the Roman Empire’s collapse a second time under laboratory conditions to isolate the variable.

Condition 2 — practical judgment remains necessary despite irreducible uncertainty. Firms, research institutions, and governments already make continuity-capital decisions daily under exactly this condition — whether to preserve a retiring engineer’s tacit process knowledge, whether a merger’s due diligence should weight institutional-memory loss, whether an R&D archive is asset or liability — without any existing framework naming what they are deciding about. The claim does not introduce uncertainty into a previously certain domain. It names uncertainty that decision-makers are already navigating unnamed.

Condition 3 — the theory specifies expected patterns before observation. Three falsifiable predictions follow directly from Papers 3 and 4, applied to this domain: continuity capital should depreciate faster in ideas whose interpretive context was never made explicit than in ideas accompanied by well-documented reasoning, independent of the ideas’ technical complexity; depreciation should show threshold, not smooth, dynamics, consistent with Paper 4; and reconstruction cost, once past that threshold, should exceed the original cost of maintaining continuity, consistent with the hysteresis Paper 4 predicts for institutional reasoning generally. These are commitments made in advance of testing them, not retrospective descriptions fitted to the Roman concrete case after the fact.

Condition 4 — the theory improves orientation among plausible interventions. A firm

or field equipped with this framework can ask a different, more actionable question than “how do we protect our IP” or “how do we retain talent”: *which of our currently non-rival, freely available ideas have continuity capital low enough that their practical value is already depreciating, undetected, toward Vitruvius’s two-thousand-year dormancy?* That question has no equivalent inside standard growth accounting, and answering it — even approximately — changes what gets invested in before the threshold in Condition 3 is crossed rather than after.

Condition 5 — the theory specifies its own conditions for revision. The claim would be weakened, and should be revised, if continuity capital turns out to correlate so tightly with existing categories — codified human capital, organizational capital, or standard measures of tacit-knowledge loss — that it adds no orientation beyond relabeling them; if the threshold dynamics predicted in Condition 3 turn out, on closer historical or experimental examination, to be smooth rather than non-linear; or if a body of counter-cases emerges in which ideas with documented, near-zero continuity capital were nonetheless activated at low cost, which would suggest the reconstruction-cost asymmetry from Paper 4 does not generalize as this paper claims.

V. Toward Measurement

Paper 5 proposed four navigational proxies for the related, narrower construct of invisible opportunity cost: reconstruction proportion, continuity depth, handoff coherence, and recovery asymmetry. Those proxies apply directly to continuity capital and require no modification — they were, in effect, already measuring this paper’s construct at organizational scale. What this paper adds is a fifth, specific to ideas considered as an aggregate stock rather than as individual organizational decisions: **continuity half-life** — the interval over which an idea’s documented existence persists unchanged while the proportion of its originating community still capable of reconstructing its underlying mechanism from first principles declines by half. This is offered as a navigational instrument in Paper 2’s sense: approximate, contestable, and useful for orientation long before it is precise enough for formal econometric estimation.

VI. Discussion

Two existing bodies of work sit close enough to this argument to require explicit positioning, beyond Romer.

Nonaka’s knowledge-creation theory, already cited elsewhere in this series for its account of tacit-to-explicit knowledge conversion (the SECI model), might seem to anticipate the present claim: Nonaka argues that organizational knowledge is created through continuous conversion between tacit and explicit forms, and that this conversion, not mere possession of information, is what generates value. The overlap is real and worth naming plainly. Where this paper departs is in claiming that the *relationship* itself — independent of any particular conversion event — is subject to depreciation dynamics of its own, with thresholds and hysteresis, in the specific sense established in Paper 4. Nonaka’s model describes how tacit knowledge becomes explicit. It does not, on its own, predict that explicit knowledge already fully externalized — Vitruvius’s text is about as explicit as a technical description gets — can

nonetheless sit economically inert for two thousand years because the community capable of reactivating it has dissolved. That is the gap this paper’s synthesis of Papers 3 and 4 fills that SECI alone does not.

Intangible capital economics (Corrado, Hulten & Sichel, already cited in Papers 1 and 5) established that organizational knowledge, brand, and process investment constitute real capital stock omitted from conventional accounting. This paper’s relationship to that literature is one of specification rather than departure: continuity capital is a candidate addition to the intangible capital stock, distinguished from its existing categories (software, R&D, organizational competencies) by attaching not to a firm’s internal investment but to the ongoing coherence between any idea, firm-internal or freely public, and whatever community retains the capacity to activate it. A firm can lose continuity capital in an idea it never owned and never internally developed, simply by losing the personnel who could still explain why a long-published method works. Existing intangible capital accounting has no category for this because its unit of analysis is the firm’s own investment, not the idea’s activation status across whatever community depends on it.

Two limitations deserve equally direct statement. First, the Roman concrete case is a single historical illustration, chosen for vividness and verifiability, not a dataset; the threshold and hysteresis predictions in Section IV require testing against cases where continuity capital can be tracked over shorter, better-documented intervals — the diffusion and subsequent loss of specific industrial or agricultural techniques within living memory would be a more tractable starting corpus than antiquity. Second, continuity half-life as proposed in Section V is a navigational instrument, not yet an operational one; specifying how “proportion of a community capable of reconstruction” would actually be estimated for a live case, rather than a settled historical one, is unfinished work this paper does not complete.

VII. Conclusion

Endogenous growth theory is correct that ideas, unlike physical capital, do not deplete with use. This paper has argued that this true and important fact has been read as implying something false: that an idea’s usefulness is guaranteed by its bare, non-rival existence, indefinitely, without further condition. Vitruvius’s concrete recipe was non-rival, freely available, and economically dormant for the entire span of recorded modern history, until three separate research efforts spent, cumulatively, over a decade reconstructing a coherent relationship to a text that had never once been lost. The idea did not depreciate. The continuity did.

This series has already established the mechanism (Paper 3), its non-linear dynamics (Paper 4), its unmeasured economic cost at organizational scale (Paper 5), and the philosophical case for treating understanding as a sustained relationship rather than a possessed object (Paper 6), against the methodological standard this series holds itself to (Paper 2) and the argument that measurement conventions are prior theoretical commitments rather than neutral discoveries (Paper 1). This paper’s contribution is to point that existing machinery at a target none of the prior six papers individually addressed: the economics of ideas themselves, at the point where growth theory currently assumes the question is already closed.

Following the categories this research programme has committed to using throughout, the

contribution of this paper is **Extension** — taking Romer’s endogenous growth theory into territory its own model does not address — built by **Synthesis** of Papers 1 through 6, with a **Proposal** attached to the terminology: continuity capital, and its associated instrument, continuity half-life. Whether the terminology survives is, as Paper 2 said of navigational validity, less important than whether the underlying claim — that the depreciation of coherent relationship to an idea is real, measurable in principle, and currently invisible to the theory that should be accounting for it — proves useful enough to test properly.

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